Disadvantages

1. Not protected well as the arrangement is in open outdoors.
2. Universal joints get damaged easily.

4.3.2 Torque tube drive

Construction

The propeller shaft of the vehicle is kept inside on another tube called as the torque tube which looks like a pipe. The construction of the Torque tube drive is shown in Figure 4.3.2.

The bearing which is connected to this allows the propeller shaft to rotate easily. Only one universal joint is fitted in this drive. The front part of the torque tube is like a cone and is connected at the back side of the gear box casing by using a ball and socket joint.

Working Principle

When the vehicle moves on the road, the torque and the drive thrust are transmitted to the chassis frame through axle casing. Since the ball and socket joint in the torque tube is supported by the roller bearing, the propeller shafts rotates easily. This torque tube drive is mostly used in cars and in heavy commercial vehicles.

Advantages

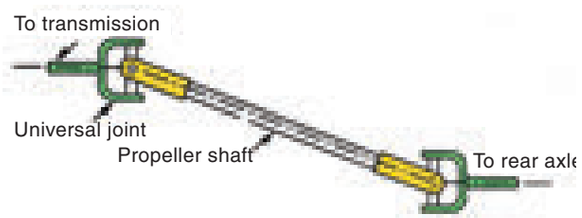
1. It's very safe to use.
2. It works for a long time.
3. Universal joint in this drive will not damage quickly

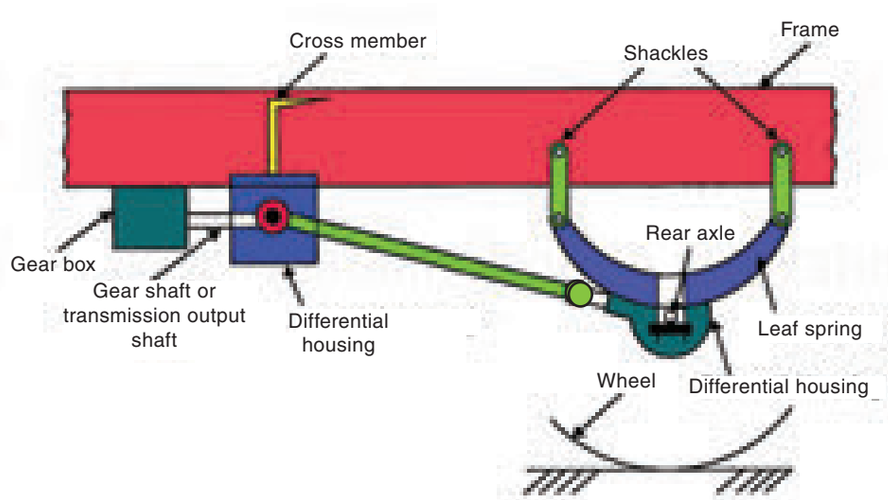
Disadvantages

1. Overweight.
2. It is not very strong.
3. It is not suitable for transmitting more power.

4.4 Parts of the propeller shaft

1. Universal joint
2. Sliding joint





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Figure 4.3.2 Torque tube drive

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